

# MINGYU KIM

## CONTACT INFORMATION

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**Email:** [k0136000@mju.ac.kr](mailto:k0136000@mju.ac.kr)

**Website:** <https://kminsalgorithm.github.io>

## RESEARCH INTEREST

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My master's research focused on enhancing dynamic obstacle avoidance performance in mobile robot navigation by leveraging deep learning models to process semantic information and generate human-like trajectories based on imitation learning.

My primary research interests are focused on:

- Learning-based path planning
- Imitation learning
- Reinforcement learning

## EDUCATIONAL QUALIFICATION

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**Myongji University**

M.S. in Information and Communication Engineering  
Total GPA of 4.5/4.5

Yongin, Korea  
Mar. 2024 - Current

**Myongji University**

B.S. in Information and Communication Engineering  
Total GPA of 3.5/4.5, Major GPA of 3.77/4.5

Yongin, Korea  
Mar. 2020 - Feb. 2024

## PUBLICATIONS

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### International Journals

- **HADP: Hybrid A\*-Diffusion Planner for Robust Navigation in Dynamic Obstacle Environments**  
M. Kim\*, C. Heo\*, J. Jung (\*co-first authors)  
*IEEE Access [Impact Factor: 3.6]*  
[read more](#)
- **HiMSELF: A Hierarchical Misbehavior Classification with Sequence Embedding by Latent Features in Vehicular Ad-Hoc Networks**  
M. Kim, D. Yum, J. Jung  
*IEEE Access [Impact Factor: 3.6]*  
[read more](#)

## Domestic Journals

- **Effective Embedding Techniques for Misbehavior Classification in Vehicular Ad-Hoc Networks**  
M. Kim, J. Jung  
Journal of Korean Institute of Information Scientists and Engineers, (KIISE 2024).

## Domestic Conferences

- **Label Similarity Analysis Based on LSTM in Vehicular Ad-hoc Networks**  
M. Kim, J. Jung  
Korea Software Congress, (KSC 2023) – **Best Paper Award**.
- **Reinforcement Learning-Based Navigation System Without Dependence on SLAM**  
M. Kim, J. Jung  
Korea Institute of Next Generation Computing, (KINGPC 2024).

## PATENTS

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- **Title: Method for producing dynamic obstacle avoidance path using conditional diffusion model**  
**Status:** Patent Application Filed  
**Application No.:** 10-2025-0053197  
**Filing Date:** Apr 23, 2025  
**Applicants:** Jahee Jung, **Mingyu Kim**, Chanyeong Heo
- **Title: System for producing dynamic obstacle avoidance path using conditional diffusion model**  
**Status:** Patent Application Filed  
**Application No.:** 10-2025-0053198  
**Filing Date:** Apr 23, 2025  
**Applicants:** Jahee Jung, **Mingyu Kim**, Chanyeong Heo

## AWARDS

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|---------------------------------------------------------------------------------------------|------|
| Encouragement Prize, Myongji University Software Competition                                | 2022 |
| Grand Prize, Myongji University Software Competition                                        | 2023 |
| Best Paper Award, Korea Software Congress (KSC)                                             | 2023 |
| Gold Prize in the Contribution Award, Myongji University                                    | 2024 |
| Awarded the Research Scholarship for M.S. Studies by the National Research Foundation (NRF) | 2024 |

## SKILLS

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### Programming Language

- Python (PyTorch, Tensorflow, FastAPI), C (MCU programming), JavaScript (NestJS)

### Machine Learning

- Imitation learning, Anomaly Detection

### Robot Development and Implementation

- Learning-Based Path Planning

- Development of a [Swerve-Drive Robot](#)
- Development of an [Omni-Wheel-Based Robot](#)

#### Development Tools

- ROS, Gazebo, Fusion 360

### RESEARCH PROJECTS

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#### Development of a Cybersecurity Platform for V2X-based Remote-Assisted Mobility

AutoCrypt

2023 – Current

#### Semantic Autonomous Driving Systems Utilizing Diffusion Models for Enhanced Dynamic Obstacle Avoidance Performance

National Research Foundation of Korea (NRF)

2024 – Current

### EXTRACURRICULAR ACTIVITIES:

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#### Undergraduate Research Intern, DS LAB (Advisor: [Jaehee Jung](#))

2022 – 2024

- Researched on improving misbehavior classification in V2X messages by analyzing relationships between misbehavior types.
- Researched on DRL-based navigation systems that operate without reliance on maps.

#### LikeLion, Myongji University

2023 – 2024

- Backend Operator & Instructor

#### AUTURBO

2024 – 2025

- Member, Outdoor Delivery Robot Team

### PERSONAL INFORMATION:

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**Date of Birth**      20-02-2001

**Hobbies**            3D Modeling  
Freelance work via [Soomgo](#)